

National School of Artificial Intelligence

Artificial Intelligence & Data Sciences

3rd Year

Project Management

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“SmartAttend: Planning an AI-Based Attendance System”

1. Case Overview

In February 2026, **Mediterranean Institute of Technology (MITech)**, a mid-sized private university located in North Africa, decided to modernize its attendance management system.

The university hosts:

- 4,800 students,
- 220 faculty members,
- 6 academic departments,
- 30 classrooms.

Attendance is currently managed manually, resulting in:

- Time loss (10–15 minutes per class)
- Proxy attendance (students signing for friends)
- Data entry errors
- Administrative delays in report generation

The Rector requested the development of a Smart AI-Based Attendance System (**SmartAttend**) to:

- Automatically recognize students using facial recognition,
- Generate real-time attendance reports,
- Integrate with the existing Learning Management System (LMS),
- Be deployed within 4 months,
- Respect student data privacy regulations.

However, the university allocated a **maximum budget of 1,200,000 DZD**.

A student AI engineering team has been invited to submit a project plan and budget proposal.

2. Technical Context Provided to the Team

To support planning decisions, the university provided the following information:

Infrastructure Status

- Classrooms already have Wi-Fi.
- No cameras are installed.
- University servers have limited capacity.
- Cloud hosting is allowed but must comply with data protection rules.

Data Considerations

- The university can provide student ID photos.
- Photos are stored in low resolution.
- Students must give consent before biometric use.

Regulatory Constraint

Under national digital data protection guidelines:

- Biometric data must be encrypted.
- Sensitive data cannot be stored on foreign servers without authorization.

3. Technical Development Options

The team must choose between two approaches:

Option A: Custom Model Development

- Develop facial recognition model from scratch.
- Higher flexibility.
- Requires large dataset and GPU usage.
- Estimated development time: 4–6 months.

- Higher cost.

Option B : Pre-Trained Model Integration

- Use an existing open-source facial recognition model.
- Fine-tune using university dataset.
- Faster deployment.
- Lower development cost.
- Limited customization.

4. Cost Reference Estimates

The university provides approximate cost assumptions:

Human Resources (monthly estimates)

- AI Engineer: 180,000 DZD
- Backend Developer: 150,000 DZD
- Frontend Developer: 130,000 DZD
- Project Manager: 160,000 DZD

Hardware Costs

- Smart classroom camera: 25,000 DZD per classroom
- GPU rental (cloud): 120,000 DZD per month

Security Add-On

- Data encryption module implementation: 200,000 DZD

Testing and integration: 150,000 DZD

5. The Planning Challenge

The team must:

1. Define project scope
2. Identify deliverable
3. Determine required resources
4. Estimate total project cost
5. Decide technical development approach
6. Prepare justification for budget

6. The Negotiation Trigger

After reviewing the team's proposal, the university administration responds:

“Your estimated budget is 4,100,000 DZD.
Our maximum budget is 1,200,000 DZD.
We cannot exceed this limit.”

The team must now renegotiate the project plan.

7. Unexpected Development

Two weeks later, a new national directive states:

“All biometric data must be stored locally on secured institutional servers.”

This may:

- Increase infrastructure cost
- Reduce cloud feasibility
- Delay deployment

The team must reassess scope, cost, and timeline.